



SNG BUILDER SERIES — GUIDE #001 OF 4

SUR-RON LBX FRAME-UP SUPER BUILD

START WITH NOTHING. END WITH A LUXURY ELECTRIC MACHINE.

\$14,714 TOTAL PARTS COST	LBX FRAME PLATFORM	\$0 DONOR BIKE NEEDED	26+ PAGES / STEPS
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No donor bike. No base to start from. Just an LBX frame and this guide. Every part is sourced, priced, and documented — and the finished machine is something you can't buy off a showroom floor at any price.

WHAT'S INSIDE THIS GUIDE

- Frame Selection & Prep**
Sur-Ron LBX frame sourcing, inspection checklist, frame prep before any parts go on
- Full Sourced Parts List**
Every component needed to build from zero — sourced, priced, and numbered. \$14,714 total.
- Motor & Drivetrain**
Motor selection, gearing, jackshaft setup, chain routing for the LBX platform
- Battery & Electrical System**
Cell selection, BMS wiring, full harness schematic, controller pairing
- Suspension & Chassis**
Fork selection, rear suspension setup, geometry notes for tall rider configuration
- Body & Finishing**
Panel selection, mounting, finishing details that separate a luxury build from a budget build
- Assembly Sequence**
Step-by-step build order with torque specs, critical checkpoints, and builder's notes
- First Start & Setup**
Power-on checklist, controller baseline programming, first ride prep

BUILDER'S NOTE — BRYAN JAWORSKI, SNG CYCLES

The LBX platform is where SNG started. This is the frame-up guide I wish existed — every sourcing decision, every mistake, every "why didn't anyone tell me this" moment is documented here. You're not buying a kit. You're building something custom, and this guide makes sure you build it right.



BUILD OVERVIEW

\$0

DONOR BIKE
REQUIRED

LBX

FRAME PLATFORM

\$14,714

ALL-IN PARTS
BUDGET

Luxury

BUILD GRADE

PLATFORM	
Base Frame	Sur-Ron LBX — sourced new or used. No donor bike needed.
Build Type	Frame-up from zero. Every component is chosen and installed per guide.
Build Grade	Luxury — premium component selection throughout. Not a budget build.
WHAT YOU START WITH	
Starting Point	Sur-Ron LBX frame (bare) — that's it. Everything else comes from the parts list.
Frame Cost	Included in the \$14,714 total
Donor Bike Needed?	No. Start from a bare LBX frame and build up.
BUILD ECONOMICS	
Total Parts Cost	\$14,714 — fully sourced and priced in the guide
Comparable Retail Value	A comparable luxury electric build isn't available retail at this spec
Build Time	Experienced builder: 80-100 hrs. First-time builder: 120-150 hrs with this guide.
Tools Required	Standard shop tools — full list in guide. No specialized equipment required.

PARTS LIST — BY CATEGORY

NOTE ON THIS INDEX

This is the category overview. The complete guide includes every individual part number, exact sourcing link or search term, current pricing, and quantity. The full sourced list is what makes this guide worth \$149 — it took 14 months to build.

CATEGORY / COMPONENT	ITEMS	NOTES	BUDGET RANGE
FRAME & CHASSIS			
Sur-Ron LBX Frame	1	New or used — condition spec in guide	See guide
Frame Prep / Hardware	Kit	Fastener kit, thread inserts, prep spec	See guide
SUSPENSION			
Front Fork Assembly	1 set	Upgrade spec — LBX platform	See guide
Rear Shock	1	Spec and setup in guide	See guide
Linkage / Bearings	1 set	Included in suspension section	See guide
MOTOR & DRIVETRAIN			
Motor	1	Full spec and supplier in guide	See guide
Gearing / Sprocket Set	1 set	Ratio spec per guide	See guide
Chain(s)	Per spec	Type, length, and master links	See guide
Jackshaft (if applicable)	1	See Guide #004 for full fab detail	~\$200-400
ELECTRICAL SYSTEM			
Controller	1	Matched to motor — programming in guide	See guide
Battery Cells + BMS	Per spec	Cell type, voltage, capacity — full spec	See guide
Charger	1	On-board or external — spec in guide	See guide
Wiring Harness	Custom	Full schematic included	See guide
Throttle / Controls	1 set	Type and calibration in guide	See guide
BODY & FINISHING			
Body Panels / Plastics	1 set	Luxury spec — sourcing in guide	See guide
Seat	1	Custom or OEM — spec and mounting in guide	See guide
Lighting / Display	Per spec	Luxury build finishing items in guide	See guide
WHEELS & BRAKES			
Wheel Set	F+R	Spec per LBX platform guide	See guide
Brakes	1 set	Hydraulic — spec in guide	See guide
Tires	F+R	Recommendations in guide	See guide
TOTAL — ALL-IN PARTS BUILD			



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This guide is sold as educational reference material. SNG Cycles / CCM Inc. assumes no liability for outcomes of builds performed using this guide. Electric vehicle builds involve high-voltage systems. Follow all applicable safety regulations.

GET THE FULL GUIDE

This document is the **product overview and parts index**. The complete Guide #001 includes all sourced part numbers, exact assembly steps, wiring diagrams, torque specs, and 14 months of builder's notes from the SNG LBX build program.

\$149

GUIDE #001 — LBX
FRAME-UP

\$149

GUIDE #002 — 270-E
BUILD

\$249

GUIDE #003 — 52KW
CRF450

\$399

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HOW TO ORDER

Email **sharknotgoldfish@gmail.com** with the guide number(s) you want. We'll send a payment link (Venmo / PayPal / Zelle) and the PDF guide within 24 hours of payment confirmation.

ABOUT SNG CYCLES

SNG Cycles is a custom electric motorcycle brand founded by Bryan Jaworski in Royal Palm Beach, Florida. Every build in the SNG lineup came out of a garage, not a boardroom. Three R&D prototypes. Fourteen months. Over \$50K of personal investment. These guides exist because the information didn't exist anywhere else — so the builder documented everything.

Shark Not Goldfish. Leave the School Behind.

BRYAN JAWORSKI

FOUNDER — SNG CYCLES / CCM INC.

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SNG BUILDER SERIES

GUIDE #001 OF 4

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SNG BUILDER SERIES — GUIDE #002 OF 4

SNG 270-E LIGHT BEE BUILD

ALREADY HAVE A LIGHT BEE? OR STARTING FRESH? THIS GUIDE COVERS BOTH.

\$16,430 TOTAL (INCL. LIGHT BEE)	~\$12,930 ADD-ON PARTS COST	Sur-Ron BASE PLATFORM	270-E SNG PROTOTYPE SPEC
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Already own a Light Bee? You're \$3,500 ahead. Don't have one? Source it and everything else — this guide has you covered either way. The result is the SNG 270-E prototype spec: a machine that doesn't exist on any showroom floor.

TWO PATHS. ONE GUIDE.

Path A — Already Have a Light Bee
Upgrade Path

Your Light Bee becomes the foundation. Skip the \$3,500 bike cost — go straight to the upgrade parts list. The guide walks you through what comes off and what goes on, in what order.

~\$12,930
in add-on parts

Path B — Starting From Zero
Full Build Path

Source the Light Bee first (new or used), then follow the same guide. Every step is documented. The \$3,500 Light Bee is just the starting point — everything else in the guide is the same.

\$16,430
all-in total

WHAT'S INSIDE THIS GUIDE

- Light Bee Platform Overview**
What comes stock, what comes off, what stays — prep checklist for both upgrade and fresh build
- Full Parts List — Sourced & Priced**
Every upgrade component for the 270-E spec. Sourcing strategy, part numbers, estimated cost.
- Drivetrain Upgrade**
Motor, controller, gearing — the 270-E spec upgrade from stock LBX drivetrain
- Battery System Upgrade**
Cell upgrade, BMS, new wiring to support increased power
- Suspension & Chassis**
Fork upgrade, geometry notes, tall rider configuration built into the SNG 270-E spec
- Body & Panel Work**
The body mods that make the 270-E look like the SNG prototype — not a stock Light Bee
- Wiring & Controls**
Full harness schematic for the upgraded system — throttle, display, cut-offs
- First Start & Tuning**
Controller baseline, post-upgrade checklist, power-on sequence

BUILDER'S NOTE — BRYAN JAWORSKI, SNG CYCLES

The 270-E is what the Light Bee becomes when somebody who actually builds takes it apart and puts it back together better. This guide is the documented build log of the SNG prototype. If you already have a Light Bee, this is the upgrade path. If you don't, source one and start here — same result either way.



BUILD ECONOMICS

\$3,500 SUR-RON LIGHT BEE (BASE)	\$12,930 UPGRADE PARTS	\$16,430 ALL-IN IF STARTING FRESH	270-E SNG PROTOTYPE SPEC
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PLATFORM & STARTING POINT	
Base Platform	Sur-Ron Light Bee (LBX)
Have a Light Bee already?	Start at the upgrade parts list (~\$12,930). Light Bee cost is already sunk.
Don't have a Light Bee?	Source one new or used (~\$3,500), then follow the same guide. Total: ~\$16,430.
Build Result	SNG 270-E prototype specification — the actual SNG build

BUILD REQUIREMENTS	
Tools Required	Standard shop tools — full list in guide. No CNC or specialized equipment.
Skill Level	Intermediate mechanical. Electrical wiring comfort required.
Build Time	60–80 hrs from base Light Bee. 100+ hrs from zero.

COST BREAKDOWN (ALL-IN PATH)	
Sur-Ron Light Bee	~\$3,500 (new or used — condition and year spec in guide)
Upgrade Components	~\$12,930 — fully sourced and priced in guide
Total	\$16,430

WHY THE SNG 270-E SPEC?

The Light Bee is one of the best platforms in the small electric space — but it's a starting point, not a finish line. The SNG 270-E is what you get when a builder who has spent 14 months and \$50K+ developing electric vehicles takes that platform apart and specs every upgrade that actually matters.

This isn't a list of bolt-on parts. It's a documented prototype build — the real SNG 270-E spec, built and tested. The guide is the blueprint for replicating it.

ALREADY IN THE SUR-RON COMMUNITY?

You've seen builds. You've seen the good ones and the overpriced ones. The 270-E spec gets you to the top tier of Sur-Ron builds for a documented cost, with every sourcing decision explained. No guessing. No "hope this part works." It works — it's on the prototype.

TERMS & HOW TO ORDER

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\$149 GUIDE #001 — LBX FRAME-UP	\$149 GUIDE #002 — 270-E BUILD	\$249 GUIDE #003 — 52KW CRF450	\$399 ALL 4 GUIDES BUNDLE
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GUIDE #002 OF 4
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SNG BUILDER SERIES — GUIDE #004 OF 4

E-520 JACKSHAFT FABRICATION & ASSEMBLY

DUAL-STAGE 15.44:1 REDUCTION SYSTEM — NO MACHINIST SKILLS REQUIRED

15.44:1 TOTAL REDUCTION	2- Stage DRIVE SYSTEM	16 Parts EXPLODED BOM	6 Steps ASSEMBLY SEQUENCE	\$ GU PR
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You don't need to be a machinist. Hand the instructions in this guide to any machine shop and they will have zero follow-up questions. Every dimension, every tolerance, every material spec is documented. The plate and jackshaft shaft are the only custom parts — everything else ships off a shelf.

WHAT'S INSIDE THIS GUIDE

- System Overview**
Dual-stage reduction concept, why 15.44:1, where this system fits in the SNG builds
- Machinist Instructions**
Ready-to-hand-over spec sheet for plate and shaft fabrication. Machinist needs nothing else.
- 16-Part Exploded BOM**
Every component numbered, sourced, and priced. What you buy vs. what gets machined.
- Critical Dimensions**
Shaft diameter, bearing spec, plate dimensions, bearing centers — all documented
- 6-Step Assembly Sequence**
Step-by-step with photos. Critical: secondary (rear) chain aligned first — always.
- Torque Specs**
Every fastener torqued to spec. M8, M10, and set screws — no guessing.
- Chain Alignment Procedure**
The sequence that matters. Why rear chain goes first. How to set both stages correctly.
- Installation Notes**
Integration into the full build — connects with Guide #001, #002, and #003 builds

BUILDER'S NOTE — BRYAN JAWORSKI, SNG CYCLES

I'm not a machinist either. That's why I wrote these instructions the way I did — so I could hand them to a machine shop and not get a single call back with questions. The plate and shaft are the only parts that need machining. Everything else — bearings, sprockets, chain, fasteners — comes off a shelf. This guide handles all of it.



REDUCTION SYSTEM — DUAL STAGE

E-520 Jackshaft — Drive Ratio Breakdown
Primary Stage
12T Drive Sprocket → 48T Driven Sprocket
4.00:1
Secondary Stage
14T Drive Sprocket → 54T Driven Sprocket
3.86:1

Total Reduction
4.00 × 3.86 — Dual-stage compound reduction
15.44:1

CRITICAL DIMENSIONS

20mm
Jackshaft Diameter
UCP204
Bearing Type
197mm
Bearing Center-to-Center
3/8"
Plate Thickness
A36/1018
Plate Material (Mild Steel)
#420
Chain Type

TORQUE SPECIFICATIONS

22 Nm
M8 Fasteners
40–45 Nm
M10 Fasteners
10–12 Nm
Set Screws

WHAT NEEDS MACHINING VS. WHAT YOU BUY

THE KEY POINT

Everything else — bearings, sprockets, chain, fasteners, hardware — is off-the-shelf. The guide includes the exact machinist instructions for both custom parts. Hand them to any machine shop. No follow-up questions. No back-and-forth.

COMPONENT	HOW TO GET IT
Mounting Plate — 3/8" Mild Steel A36/1018	Machine shop — instructions in guide. Exact dimensions, hole pattern, and spec sheet ready to hand over.
Jackshaft — 20mm	Machine shop — instructions in guide. Machinist gets the drawing and asks no further questions.
UCP204 Pillow Block Bearings	Off the shelf — eBay / bearing supplier
Primary Sprockets (12T + 48T)	Off the shelf — eBay / sprocket supplier
Secondary Sprockets (14T + 54T)	Off the shelf — eBay / sprocket supplier
#420 Chain (Primary)	Off the shelf — length in guide
Chain (Secondary/Final Drive)	Off the shelf — per guide spec
Fastener Kit (M8 + M10 + set screws)	Off the shelf — hardware store or eBay

6-STEP ASSEMBLY SEQUENCE

- 1
- Receive Machined Parts — Inspect Before Assembly
Verify plate dimensions and shaft diameter before doing anything else. Check bearing bore fit on shaft. Full inspection checklist in guide.
- 2
- Mount UCP204 Bearings to Plate
197mm center-to-center. Torque bearing mount hardware to spec. Do not fully tighten until shaft is installed and both chains are set.
⚠ Do not fully tighten bearing mounts at this stage.
- 3
- Install Jackshaft Through Bearings
Slide 20mm shaft through both UCP204 bearing bores. Stage your sprockets on the shaft in correct position before proceeding — position spec in guide.
- 4
- Set Secondary (Rear) Chain First — Always
Align the 14T→54T secondary stage chain before touching the primary. This is the critical step. Chain alignment sequence is documented in detail in the guide.
⚠ Critical: Secondary chain alignment first — every time. Primary chain alignment depends on this position.
- 5
- Set Primary Chain — Then Lock Everything
With secondary set, align primary 12T→48T chain. When both chains are running true with correct tension, fully torque all bearing mounts and set screws to spec.
M8: 22 Nm · M10: 40–45 Nm · Set Screws: 10–12 Nm
- 6
- Final Inspection & Installation
Rotation check, chain tension verify, torque confirmation. Mount to frame per build guide (compatible with Guide #001, #002, and #003 builds). Full installation notes in guide.



ADDITIONAL NOTES — WP SUSPENSION & BRAKES

WP FORK CONVERSION — LEWIS BRAKES ADAPTER

When the SNG builds were in development, no ready-made adapter existed for WP fork conversion with the Lewis Brakes system. SNG developed CAD drawings for the adapter at that time. Since then, Lewis Brakes has released a complete kit at \$249 — and this is now the recommended path. Even if you fabricated the adapter yourself, you'd still need a special notched axle and different spacer. The math doesn't work for DIY. Buy the Lewis Brakes kit.

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GET THE FULL GUIDE

This document is the **product overview and technical index**. The complete Guide #004 includes the machinist-ready fabrication spec sheet, full exploded BOM with part numbers, assembly photos for all 6 steps, and installation notes for each SNG build platform.

\$49

GUIDE #004 — JACKSHAFT

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GUIDE #001 — LBX FRAME-UP

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GUIDE #002 — 270-E BUILD

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